Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn about. You have chosen the topic of:

Nuclear Fusion

Nuclear fusion is the process of combining two or more atomic nuclei into a single, heavier nucleus, releasing a large amount of energy in the process. Nuclear fusion is the source of energy for the Sun and other stars.

Some of the main points to know about nuclear fusion are:

- Nuclear fusion requires high temperatures and pressures to overcome the electrostatic repulsion between positively charged nuclei. The conditions for fusion are achieved in the cores of stars, where the temperature is about 15 million degrees Celsius and the pressure is about 250 billion atmospheres.
- The most common fusion reaction in stars is the proton-proton chain, which converts four hydrogen nuclei (protons) into one helium nucleus (alpha particle), releasing two positrons, two neutrinos, and gamma rays. The mass of the helium nucleus is slightly less than the mass of the four protons, and the difference is converted into energy according to Einstein's equation E = mc^2.
- Another fusion reaction that occurs in stars is the carbon-nitrogen-oxygen (CNO) cycle, which uses carbon, nitrogen, and oxygen as catalysts to fuse hydrogen into helium. The CNO cycle is dominant in stars that are more massive and hotter than the Sun.
- Nuclear fusion can also be achieved artificially in devices called fusion reactors, which use magnetic fields or lasers to confine and heat plasma, a state of matter where atoms are ionized. The most common fusion reaction in fusion reactors is the deuterium-tritium (D-T) reaction, which fuses one deuterium nucleus (an isotope of hydrogen with one proton and one neutron) and one tritium nucleus (an isotope of hydrogen with one proton and two neutrons) into one helium nucleus and one neutron, releasing a lot of energy.
- Nuclear fusion has many potential advantages as a source of energy, such as being abundant, clean, safe, and sustainable. However, there are also many challenges and difficulties in achieving and maintaining fusion, such as the high cost, complexity, and technical feasibility of fusion reactors, the environmental and security risks of handling radioactive materials, and the ethical and political implications of nuclear technology.

Workshop Practice Lab:

- Workshop practice lab is a common theory/lab course for all engineering students in the first and second semester .

- The objective of this lab is to get a hands-on knowledge of several workshop practices like carpentry, fitting, welding, machining, etc. and learn safety regulations to be maintained in a shop floor.
- Workshop practice is the backbone of the real industrial environment which helps to develop and enhance relevant technical hand skills required by the technician working in the various engineering industries and workshops.
- The lab syllabus may vary depending on the university or college, but some of the common experiments are :
  - Introduction to workshop tools and safety precautions
  - Carpentry: making various joints and patterns using wood and nails
  - Fitting: filing, sawing, drilling, tapping, and making various joints using metal pieces
  - Welding: arc welding, gas welding, and spot welding of metal plates and rods
  - Machining: turning, milling, shaping, and grinding of metal workpieces using lathe, milling machine, shaper, and grinder
  - Sheet metal: cutting, bending, and forming of sheet metal using hand tools and machines
  - Plumbing: cutting, threading, and joining of pipes and fittings
  - Electrical and electronics: wiring, soldering, and testing of electrical and electronic circuits and components
- The lab work consists of four parts :
  - A pre-lab homework that the students must complete before coming to the lab
  - The lab itself, answering all the warm-ups and predictions, and attaching data, results, graphs, and analysis
  - A post-lab report that the students must submit after the lab
  - A viva or oral examination that the students must attend to demonstrate their understanding of the lab concepts and skills
- The lab work is evaluated based on the following criteria :
  - Attendance and participation
  - Pre-lab homework
  - Lab performance and results
  - Post-lab report
  - Viva or oral examination
- The lab work is expected to develop the following outcomes in the students:
  - Ability to use various hand tools and machines in a safe and efficient manner
  - Ability to perform various workshop operations and processes with accuracy and precision
  - Ability to measure and inspect the workpieces using appropriate instruments and methods
  - Ability to analyze and interpret the data and results obtained from the experiments
  - Ability to communicate and document the lab work effectively and

professionally

S. No. Mechanical Workshop Duration

- S. No. stands for serial number, which is a way of numbering items in a list or a table.
- Mechanical workshop is a practical course that involves learning and applying various skills and techniques related to mechanical engineering, such as machining, welding, fitting, carpentry, etc.
- Duration is the amount of time that a workshop lasts, usually measured in hours or days.
- The table below shows an example of how to write the S. No., mechanical workshop, and duration for a hypothetical curriculum.

S. No.	Mechanical Workshop	Duration
1	Machining	40 hours
2	Welding	30 hours
3	Fitting	20 hours
4	Carpentry	15 hours
5	Sheet Metal Work	10 hours

1 Introduction to Mechanical workshop material, tools and machines 3 Hrs

- Mechanical workshop is a place where various materials, tools and machines are used to perform different operations such as cutting, shaping, drilling, welding, etc.
- The main objectives of mechanical workshop are to provide practical knowledge and skills to the students, to familiarize them with the common tools and machines used in engineering, and to develop their creativity and problem-solving abilities.
- The basic materials used in mechanical workshop are metals, alloys, plastics, wood, etc. Each material has its own properties, advantages and disadvantages, and applications.
- The common tools used in mechanical workshop are classified into measuring tools, marking tools, cutting tools, holding tools, striking tools, and finishing tools. Some examples of these tools are ruler, caliper, scriber, hacksaw, vise, hammer, file, etc.
- The common machines used in mechanical workshop are classified into power-driven machines and manually operated machines. Some examples of these machines are lathe, milling machine, drilling machine, grinding machine, etc.
- The safety rules and precautions to be followed in mechanical workshop are to wear proper protective equipment, to handle the tools and machines carefully, to avoid loose clothing and jewelry, to keep the work area clean

and organized, and to report any accidents or injuries immediately.

To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc) used in workshop

- Workshop layout is the arrangement of the machinery and equipment in a workshop, which affects the efficiency, quality, and safety of the work process.
- There are three basic types of workshop layout: process layout, product layout, and fixed-position layout. A process layout groups machines and equipment according to their functions, such as cutting, drilling, or welding. A product layout arranges machines and equipment in a sequence that follows the production flow of a specific product or a group of similar products. A fixed-position layout keeps the product in one place and brings the machines and equipment to it.
- A good workshop layout should consider the following factors: the space required for each machine and equipment, the space required for allied equipment (such as dust collectors, heating or cooling systems, exhaust fans, etc.), the space required for employee welfare (such as coffee machine, drinking water, etc.), the space required for gangways and emergency exits, the power supply and distribution, the lighting and ventilation, the noise and vibration control, the material handling and storage, and the safety and health standards.
- Workshop safety is the set of rules and practices that aim to prevent accidents, injuries, and hazards in the workshop. Some of the general workshop safety rules are :
  - Always wear safety gear, such as gloves, goggles, ear plugs, helmets, etc., while working in the workshop.
  - Do not wear loose, baggy, or dangling clothing, jewelry, or hair that can get caught in the machines or equipment.
  - Do not operate any machine or equipment without proper training, authorization, and supervision.
  - Do not use any machine or equipment that is defective, damaged, or missing safety guards or devices.
  - Do not leave any machine or equipment running unattended or in an unsafe condition.
  - Do not overload, over speed, or force any machine or equipment beyond its capacity or limits.
  - Do not touch any moving parts or hot surfaces of any machine or equipment.
  - Do not use any machine or equipment for a purpose other than its intended use.
  - Do not use any flammable, explosive, or hazardous materials without proper precautions and ventilation.
  - Do not smoke, eat, or drink in the workshop.

- Keep the workshop clean, tidy, and free of clutter, spills, and obstructions.
  - Store and dispose of all materials, tools, and waste properly and safely.
  - Report any accidents, injuries, or hazards to the supervisor or the authority immediately.
  - Follow the emergency procedures and instructions in case of fire, power failure, or any other emergency.
- Engineering materials are the substances that are used to make or modify the machines and equipment in the workshop. Some of the common engineering materials used in workshop are:
  - Mild steel: a low-carbon steel that is easy to cut, weld, and form. It has good strength, ductility, and toughness, but low hardness and wear resistance. It is used for making structural components, pipes, bolts, nuts, etc.
  - Medium carbon steel: a steel that has a higher carbon content than mild steel, which increases its strength, hardness, and wear resistance, but reduces its ductility and weldability. It is used for making gears, shafts, axles, springs, etc.
  - High carbon steel: a steel that has a very high carbon content, which makes it very hard, strong, and brittle, but difficult to cut, weld, and form. It is used for making cutting tools, knives, blades, etc.
  - High speed steel: a steel that has a high alloy content, which gives it high hardness, strength, and wear resistance, even at high temperatures and speeds. It is used for making drill bits, taps, dies, etc.
  - Cast iron: a iron that has a high carbon content, which makes it hard, brittle, and resistant to wear and corrosion, but difficult to weld and form. It is used for making machine frames, engine blocks, cylinders, etc.

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.

- Fitting is the process of assembling, adjusting, and repairing parts or components to make them fit together properly and function well.
- Sheet metal is a thin and flat piece of metal that can be cut, bent, and shaped into various forms and products.
- Welding is the process of joining two or more pieces of metal by applying heat, pressure, or both, to create a strong and permanent bond.

Some of the common tools, equipment, devices, and machines used in these processes are:

- **Fitting tools**: These are used to measure, mark, cut, file, drill, tap, ream, and fit parts or components. Some examples are:
  - Steel rule: A measuring device that has graduations in millimeters and inches.

- Scriber: A sharp-pointed tool that is used to mark lines or points on metal surfaces.
- Centre punch: A tool that is used to make small indentations on metal surfaces to guide the drill bit.
- Hammer: A tool that is used to strike or drive nails, pins, chisels, etc.
- Chisel: A tool that has a sharp edge and is used to cut or shape metal by striking with a hammer.
- File: A tool that has a rough surface and is used to smooth, shape, or remove metal by rubbing.
- Hacksaw: A saw that has a fine-toothed blade and is used to cut metal by moving back and forth.
- Drill: A machine that has a rotating bit and is used to make holes in metal by applying pressure.
- Tap: A tool that has a threaded end and is used to cut internal threads in holes by turning.
- Die: A tool that has a threaded hole and is used to cut external threads on rods or bolts by turning.
- Reamer: A tool that has a tapered end and is used to enlarge or finish holes by rotating.

- **Sheet metal tools**: These are used to cut, bend, shape, and join sheet metal. Some examples are:
  - Shears: A tool that has two blades and is used to cut sheet metal by applying force.
  - Snips: A type of shears that has curved or straight blades and is used to cut sheet metal along curves or straight lines.
  - Punch: A tool that has a sharp end and is used to make holes in sheet metal by striking with a hammer or a press.
  - Flange: A tool that has a bent end and is used to make a raised edge on sheet metal by striking with a hammer or a press.
  - Bender: A tool or a machine that is used to bend sheet metal into various angles or shapes by applying force or leverage.
  - Folder: A machine that is used to fold sheet metal along a line by applying pressure.
  - Roller: A machine that has two or more cylinders and is used to bend or curve sheet metal by passing it between them.
  - Planishing hammer: A tool or a machine that has a smooth head and is used to flatten, shape, or smooth sheet metal by striking or pressing.
- **Welding tools**: These are used to join two or more pieces of metal by applying heat, pressure, or both. Some examples are:
  - Welding torch: A device that produces a flame by burning a fuel gas and oxygen and is used to heat metal for welding.
  - Welding electrode: A metal rod or wire that is used to carry electric current and create an arc between itself and the metal to be welded.
  - Welding rod: A metal rod or wire that is used to supply filler metal

for welding.
  - Welding machine: A device that supplies electric current or gas for welding.
  - Welding helmet: A protective device that covers the head and face and has a darkened lens to shield the eyes from the bright light of welding.
  - Welding gloves: A protective device that covers the hands and wrists and protects them from heat, sparks, and electric shock.
  - Welding clamp: A device that holds two or more pieces of metal together for welding.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

- The least count of an instrument is the smallest measurement that can be taken with it. It is equal to the ratio of the main scale division to the number of divisions on the vernier scale or the micrometer scale.
- The least count of a vernier calliper is given by:
  - Least count = 1 main scale division (MSD) / Number of vernier scale divisions (VSD)
  - For example, if the main scale has 1 mm divisions and the vernier scale has 10 divisions, then the least count is $1/10 = 0.1$ mm.
- The least count of a vernier height gauge is the same as that of a vernier calliper, since both use the same principle of vernier scale. The only difference is that the height gauge measures the vertical distance of an object from a base, while the calliper measures the external or internal dimensions of an object.
- The least count of a micrometer (screw gauge) is given by:
  - Least count = Pitch of the screw / Number of divisions on the thimble
  - For example, if the pitch of the screw is 0.5 mm and the thimble has 50 divisions, then the least count is $0.5/50 = 0.01$ mm.
- To take different readings over given metallic pieces using these instruments, the following steps are followed:
  - For vernier calliper and height gauge:
    * Place the object between the jaws of the calliper or on the base of the height gauge and adjust the sliding jaw until it touches the object firmly.
    * Read the main scale reading (MSR) where the zero of the vernier scale coincides with a main scale division.
    * Read the vernier scale reading (VSR) where a vernier scale division coincides with a main scale division.

    * Calculate the total reading by adding the MSR and the product of VSR and the least count.

  - For micrometer (screw gauge):

    * Place the object between the anvil and the spindle of the micrometer and rotate the ratchet until a slight resistance is felt.
    * Read the main scale reading (MSR) where the edge of the thimble coincides with a main scale division.
    * Read the thimble scale reading (TSR) where the zero of the thimble coincides with a thimble scale division.
    * Calculate the total reading by adding the MSR and the product of TSR and the least count.

2 Machine shop 3 Hrs

- A machine shop is a place where various machines and tools are used to cut, shape, and form metal or other materials into desired shapes and sizes.

- Machine shops can perform various operations such as drilling, milling, turning, threading, tapping, grinding, honing, lapping, broaching, sawing, shearing, bending, forming, welding, and assembling.

- Machine shops can also produce custom-made parts or components for various industries such as aerospace, automotive, medical, defense, and manufacturing.

- Machine shops require skilled workers who can operate and maintain the machines and tools, as well as follow safety and quality standards.

- Machine shops can be classified into different types based on the size, complexity, and specialization of the work they do. Some examples are:

  - Job shops: These are small-scale machine shops that perform a variety of tasks for different customers, often on a one-time or short-term basis. They can handle small to medium-sized batches of parts or products, and can adapt to changing specifications and requirements.
  - Production shops: These are large-scale machine shops that produce high volumes of standardized parts or products for a specific industry or market. They can handle mass production and automation, and can achieve economies of scale and efficiency.
  - Prototype shops: These are machine shops that specialize in creating and testing new designs or concepts for parts or products. They can handle complex and innovative tasks, and can provide feedback and improvement suggestions.
  - Repair shops: These are machine shops that focus on fixing or restoring damaged or worn-out parts or products. They can handle main-

tenance and refurbishment, and can extend the service life and performance of the parts or products.

Demonstration of working, construction and accessories for Lathe machine

- A lathe machine is a machine tool that is used to perform various operations such as turning, facing, threading, knurling, parting, etc. on a workpiece by removing unwanted material in the form of chips.
- The working principle of a lathe machine is based on the relative motion between the workpiece and the cutting tool. The workpiece is held between two rigid and strong supports called centers or in a chuck or face plate which revolves. The cutting tool is rigidly held and supported in a tool post which is fed against the revolving workpiece. The tool can be moved parallel or perpendicular to the axis of the workpiece to produce different shapes and sizes.
- The construction of a lathe machine consists of the following main parts:
  - Bed: It is the base of the machine that supports all the other parts and provides rigidity and alignment. It is usually made of cast iron or steel and has guide ways or rails on which the carriage and tailstock can slide.
  - Headstock: It is the fixed part of the machine that holds the spindle and the driving mechanism. The spindle is a hollow cylindrical shaft that rotates the workpiece or the chuck. The driving mechanism consists of an electric motor, a gear box, a cone pulley, and a belt or a gear train that transmit the power and speed to the spindle.
  - Tailstock: It is the movable part of the machine that supports the other end of the workpiece. It has a quill or a barrel that can slide in and out to adjust the length of the workpiece. The quill has a tapered hole that can hold a center, a drill bit, or a reamer for various operations.
  - Carriage: It is the part of the machine that holds and moves the cutting tool along the workpiece. It consists of the following components:
    - Saddle: It is the horizontal part of the carriage that slides on the bed along the longitudinal axis of the workpiece.
    - Cross slide: It is the part of the carriage that slides on the saddle along the transverse axis of the workpiece. It has a hand wheel or a feed screw that can move the tool in or out of the workpiece.
    - Compound slide: It is the part of the carriage that slides on the cross slide along an inclined axis. It has a swivel base that can be rotated to adjust the angle of the tool for taper turning or angular cutting.
    - Tool post: It is the part of the carriage that holds the tool in a fixed position. It has a clamping device that can secure the tool or a tool holder. It can be of different types such as single,

four-way, or quick change tool post.

- Feed mechanism: It is the part of the machine that controls the movement of the carriage and the tool along the workpiece. It consists of the following components:
    * Lead screw: It is a long threaded rod that runs parallel to the bed and connects the headstock and the carriage. It is used for threading operations and can be engaged or disengaged by a half nut lever on the carriage.
    * Feed rod: It is a rod that runs parallel to the lead screw and connects the headstock and the carriage. It is used for turning and facing operations and can be engaged or disengaged by a feed clutch lever on the carriage.
    * Feed gears: They are a set of gears that connect the spindle and the feed rod or the lead screw. They can be changed to vary the feed rate or the pitch of the thread.
    * Apron: It is the part of the carriage that houses the feed mechanism and the half nut lever, the feed clutch lever, and the hand wheel for manual movement of the carriage.

- The accessories of a lathe machine are the additional parts or devices that are used to enhance the performance or the range of operations of the machine. Some of the common accessories are:
    - Chuck: It is a device that is mounted on the spindle or the tailstock to hold the workpiece. It can be of different types such as three-jaw, four-jaw, collet, or magnetic chuck.
    - Face plate: It is a flat circular plate that is mounted on the spindle to hold irregular or large workpieces that cannot be held by a chuck.
    - Steady rest: It is a device that is mounted on the bed to support long or slender workpieces that tend to bend or vibrate during machining. It has three adjustable jaws that can clamp the workpiece at a suitable position.

3 Fitting shop 3 Hrs

- Fitting shop is a workshop where metal components are fitted together by filing, chiseling, sawing, drilling, tapping, etc.
- The main tools used in fitting shop are hand tools, such as hammers, files, chisels, hacksaws, drills, taps, dies, etc.
- The main operations performed in fitting shop are:
    - Filing: It is the process of removing excess material from a workpiece by rubbing a file over its surface.
    - Chiseling: It is the process of cutting or shaping a workpiece by striking a chisel with a hammer.
    - Sawing: It is the process of cutting a workpiece into desired lengths or shapes by using a hacksaw or a power saw.
    - Drilling: It is the process of making holes in a workpiece by using a drill bit and a drilling machine.

- Tapping: It is the process of cutting internal threads in a hole by using a tap and a tap wrench.
    - Dieing: It is the process of cutting external threads on a rod or a bolt by using a die and a die holder.
- The main safety precautions to be followed in fitting shop are:
    - Wear protective gloves, goggles, apron, and shoes while working.
    - Use the right tool for the right job and keep them in good condition.
    - Clamp the workpiece securely on the workbench or the vice before performing any operation.
    - Do not use excessive force or speed while filing, chiseling, sawing, drilling, tapping, or dieing.
    - Do not leave any sharp edges or burrs on the workpiece after finishing the operation.
    - Keep the work area clean and tidy and dispose of the waste material properly.

1. **Practice marking operations.**

- Marking operations are the process of applying marks or symbols to a workpiece or a drawing to indicate dimensions, positions, directions, or other information.
- Marking operations are essential for accurate and efficient machining, fabrication, assembly, inspection, and testing of products or components.
- Marking operations can be performed manually or automatically, using various tools and methods such as pencils, scribers, punches, stamps, chalks, tapes, stickers, labels, lasers, inkjets, etc.
- Marking operations should follow some general guidelines, such as:
    - Use clear and consistent marks that are easy to read and understand by anyone involved in the process.
    - Use appropriate tools and methods that suit the material, shape, size, and surface condition of the workpiece or drawing.
    - Use standard symbols and abbreviations that conform to the relevant codes, standards, or specifications.
    - Use accurate and precise measurements and alignments that ensure the correctness and quality of the marks.
    - Use durable and permanent marks that can withstand the subsequent operations and environmental conditions.
    - Use safe and clean marks that do not damage the workpiece or drawing, or cause any health or environmental hazards.
- Marking operations should be practiced regularly to improve the skills and knowledge of the operators, and to ensure the consistency and reliability of the marks.
- Marking operations should be checked and verified before, during, and after the process, to detect and correct any errors or defects, and to ensure the compliance and conformity of the marks.

2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

- A U or V -shape male female work piece is a pair of metal pieces that fit together in a U or V shape, with one piece having a protruding part and the other having a corresponding recess.
- The purpose of making such a work piece is to demonstrate the skills of filing, sawing, drilling and grinding metal accurately and precisely.
- The steps involved in preparing a U or V -shape male female work piece are:
  - Selecting the appropriate metal material, such as mild steel, brass or aluminium, and cutting it to the required size using a hacksaw or a power saw.
  - Marking the outline of the U or V shape on both pieces of metal using a scriber, a steel rule and a try square. The outline should be symmetrical and have the same dimensions on both pieces.
  - Clamping the metal pieces securely on a bench vice or a machine vice and filing the edges and corners of the U or V shape using a flat file or a half-round file. The filing should be done in one direction and with even pressure to avoid creating burrs or scratches on the metal surface.
  - Checking the accuracy and smoothness of the filed shape using a vernier caliper, a micrometer or a surface plate. The shape should match the marked outline and have a smooth and flat surface.
  - Drilling holes on the protruding part and the recess of the U or V shape using a drill press or a hand drill. The holes should be aligned and have the same diameter and depth on both pieces. The drilling should be done with a centre punch, a drill bit and a cutting fluid to prevent the metal from overheating or cracking.
  - Grinding the edges and surfaces of the U or V shape using a bench grinder or a belt sander. The grinding should be done with a coarse or a fine abrasive wheel or belt, depending on the desired finish. The grinding should remove any burrs, marks or irregularities on the metal surface and create a smooth and shiny appearance.
  - Fitting the U or V shape male female work piece together and checking the fit and alignment using a feeler gauge, a straight edge or a dial indicator. The fit should be snug and without any gaps or overlaps. The alignment should be parallel and perpendicular to the base of the work piece.

4 Carpentry Shop 3 Hrs

- Carpentry is the skill of making and repairing wooden objects and structures, such as furniture, doors, windows, stairs, etc.

- A carpentry shop is a place where carpenters work with various tools and materials to create or fix wooden items.
- Some of the common tools used in a carpentry shop are:
  - Hammer: A tool with a metal head and a wooden handle, used to drive nails, pull nails, or break objects.
  - Saw: A tool with a sharp blade and a handle, used to cut wood or other materials.
  - Chisel: A tool with a metal blade and a wooden handle, used to carve or shape wood or other materials.
  - Plane: A tool with a flat metal blade and a wooden handle, used to smooth or level wood surfaces.
  - Drill: A tool with a rotating bit and a handle, used to make holes in wood or other materials.
  - Screwdriver: A tool with a metal tip and a handle, used to turn screws or bolts.
  - Wrench: A tool with a metal head and a handle, used to tighten or loosen nuts or bolts.
  - Pliers: A tool with two metal jaws and a handle, used to grip, bend, or cut wires or other materials.
  - Tape measure: A tool with a flexible metal or plastic strip and a case, used to measure length or distance.
  - Level: A tool with a liquid-filled tube and a bubble, used to check if a surface is horizontal or vertical.
  - Square: A tool with a metal or plastic ruler and a right angle, used to check or mark angles or corners.
  - Clamp: A tool with two metal or plastic jaws and a screw, used to hold or secure objects together.
- Some of the common materials used in a carpentry shop are:
  - Wood: A natural material obtained from trees, used to make various wooden objects and structures.
  - Plywood: A manufactured material made of thin layers of wood glued together, used to make flat or curved surfaces.
  - Particle board: A manufactured material made of wood chips or shavings glued together, used to make cheap or lightweight furniture or panels.
  - MDF: A manufactured material made of wood fibers and resin, used to make smooth or dense surfaces or shapes.
  - Hardboard: A manufactured material made of wood fibers and glue, used to make thin or rigid boards or sheets.
  - OSB: A manufactured material made of wood strands and adhesive, used to make strong or water-resistant panels or sheathing.
  - Lumber: A term for wood that has been cut and processed into beams, planks, or boards, used to make frames, floors, walls, etc.
  - Nails: Metal fasteners with a pointed end and a flat head, used to join wood or other materials together.
  - Screws: Metal fasteners with a threaded end and a slotted head, used

to join wood or other materials together.
  - Bolts: Metal fasteners with a threaded end and a hexagonal head, used to join wood or other materials together with nuts.
  - Nuts: Metal fasteners with a threaded hole and a hexagonal shape, used to secure bolts or other threaded fasteners.
  - Washers: Metal or plastic discs with a hole in the center, used to distribute pressure or prevent loosening of fasteners.
  - Glue: A sticky substance, usually liquid or gel, used to bond wood or other materials together.
  - Varnish: A clear or colored substance, usually liquid, used to coat wood or other materials to protect or enhance their appearance.
  - Paint: A colored substance, usually liquid, used to coat wood or other materials to change or improve their appearance.
  - Stain: A colored substance, usually liquid, used to penetrate wood or other materials to change or enhance their color.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

- Carpentry tools and equipment are used to measure, mark, cut, shape, join and finish wood materials for various purposes. Some of the common tools and equipment are:
  - Measuring and marking tools, such as rulers, tapes, squares, compasses, gauges, pencils, etc.
  - Cutting tools, such as saws, chisels, planes, knives, etc.
  - Shaping tools, such as rasps, files, spokeshaves, etc.
  - Joining tools, such as hammers, mallets, nails, screws, clamps, etc.
  - Finishing tools, such as sandpaper, scrapers, brushes, etc.
- Carpentry joints are the methods of connecting two or more pieces of wood together. There are many types of joints, each with its own advantages and disadvantages. Some of the common joints are:
  - Butt joint: The simplest and weakest joint, where the end of one piece is joined to the edge or face of another piece. Usually reinforced with nails, screws, or glue.
  - Miter joint: A joint where the ends of two pieces are cut at an angle, usually 45 degrees, and joined together. Used for making corners or frames. Usually reinforced with nails, screws, glue, or splines.
  - Dado joint: A joint where a groove is cut across the grain of one piece, and another piece is fitted into the groove. Used for making shelves, cabinets, or drawers. Usually reinforced with glue or nails.
  - Lap joint: A joint where two pieces overlap each other, either partially or fully. Used for making frames, boxes, or structures. Usually reinforced with nails, screws, glue, or dowels.

  - Bridle joint: A joint where a tenon (a projecting piece) is cut on the end of one piece, and a mortise (a matching hole) is cut on the end of another piece. The tenon is inserted into the mortise, forming a T-shaped connection. Used for making frames, chairs, or tables. Usually reinforced with glue or wedges.
  - Dowel joint: A joint where two pieces are aligned and joined by wooden pegs (dowels) that fit into holes drilled in both pieces. Used for making furniture, cabinets, or frames. Usually reinforced with glue.
  - Mortise and tenon joint: A joint where a tenon is cut on the end of one piece, and a mortise is cut on the edge or face of another piece. The tenon is inserted into the mortise, forming a right-angle connection. Used for making furniture, doors, or frames. Usually reinforced with glue, wedges, or pins.
  - Box joint: A joint where the ends of two pieces are cut into interlocking fingers, and joined together. Used for making boxes, drawers, or chests. Usually reinforced with glue.
  - Dovetail joint: A joint where the ends of two pieces are cut into interlocking wedges, and joined together. Used for making boxes, drawers, or chests. Usually reinforced with glue.
  - Tongue and groove joint: A joint where a tongue (a protruding edge) is cut on the edge of one piece, and a groove (a matching slot) is cut on the edge of another piece. The tongue is inserted into the groove, forming a flush connection. Used for making flooring, paneling, or siding. Usually reinforced with glue or nails.
  - Cross lap joint: A joint where two pieces cross each other at right angles, and a notch is cut on both pieces to allow them to fit together. Used for making frames, structures, or lattices. Usually reinforced with glue or nails.
  - Splice joint: A joint where two pieces are joined end to end, either by overlapping them or by inserting a third piece between them. Used for making longer pieces of wood from shorter ones. Usually reinforced with glue, nails, screws, or plates.
  - Finger joint: A joint where the ends of two pieces are cut into interlocking fingers, and joined end to end. Used for making longer pieces of wood from shorter ones. Usually reinforced with glue.
  - Birdsmouth joint: A joint where a notch is cut on the end of one piece, and another piece is fitted into the notch at an angle. Used for making rafters, trusses, or roofs. Usually reinforced with nails or screws.

5 Welding Shop 6 Hrs

- Welding is a process of joining two or more metal pieces by applying heat, pressure, or both, and with or without filler material.
- There are different types of welding processes, such as arc welding, gas

welding, resistance welding, solid state welding, and others.

- Each welding process has its own advantages, disadvantages, applications, and safety precautions.
- Some common welding equipment and tools are welding machine, electrodes, torch, gas cylinders, regulators, hoses, clamps, protective gear, etc.
- Some common welding defects are cracks, porosity, slag inclusion, incomplete fusion, distortion, spatter, undercut, etc.
- Some common welding symbols are groove weld, fillet weld, plug weld, slot weld, spot weld, seam weld, etc.
- Some common welding tests are destructive tests, such as tensile test, bend test, impact test, etc., and non-destructive tests, such as visual inspection, radiographic test, ultrasonic test, magnetic particle test, etc.

Introduction to BI standards and reading of welding drawings. Practice of Making following operations Butt Joint Lap Joint TIG Welding MIG Welding

- BI standards are the British standards for welding and fabrication, which specify the requirements for materials, design, execution, testing and inspection of welded structures and components.
- Reading of welding drawings involves interpreting the symbols, dimensions, tolerances, notes and specifications that indicate how to perform the welding operations and what the final product should look like.
- A butt joint is a type of joint where two pieces of metal are joined along a single edge or plane, usually by welding or brazing. The joint can be square, beveled, V-shaped, U-shaped or J-shaped, depending on the angle and shape of the edge preparation.
- A lap joint is a type of joint where two pieces of metal are overlapped and joined together, usually by welding, riveting or bolting. The joint can be single-lap or double-lap, depending on the number of overlapping layers.
- TIG welding, or tungsten inert gas welding, is a type of arc welding that uses a non-consumable tungsten electrode and an inert gas (usually argon) to create a weld pool and a shielding gas to protect the weld from atmospheric contamination. TIG welding can produce high-quality welds on thin or delicate materials, such as stainless steel, aluminum, copper and titanium.
- MIG welding, or metal inert gas welding, is a type of arc welding that uses a consumable wire electrode and an inert gas (usually argon or helium) to create a weld pool and a shielding gas to protect the weld from atmospheric contamination. MIG welding can produce fast and efficient welds on thick or thin materials, such as steel, aluminum, nickel and copper alloys.

6 Moulding and Casting Shop 6 Hrs

- Moulding and casting are two processes that are often used in the manufacturing of products. Both processes involve creating a negative space in order to create a positive product.
- Moulding is the process of shaping a material by using a rigid frame or model called a mould. The mould can be made of various materials such as metal, wood, plastic, or clay. The material to be moulded is usually a liquid or a pliable solid that can fill the mould cavity. Examples of moulding materials are plaster, clay, wax, resin, rubber, and silicone.
- Casting is the process of pouring a liquid material into a mould and letting it solidify. The solidified material is then removed from the mould and is called a casting. Casting can be used to make products from metals, alloys, ceramics, concrete, plaster, and other materials.
- Moulding and casting can be used for various purposes such as making sculpture, jewelry, tools, parts, and special effects. Moulding and casting can also be combined with other techniques such as carving, painting, and finishing to create more complex and realistic products .
- Moulding and casting shop is a place where moulding and casting materials and equipment are available for purchase or use. Moulding and casting shop can offer a range of products such as liquid latex, latex casting rubber, platinum silicone rubber, casting plaster, hydrocal, casting stone, plaster of Paris, lightweight casting and carving materials, graphite, steel, and cast iron molds, and kits for casting hands, bellies, and other body parts .
- Moulding and casting shop can also provide services such as custom mould making, casting, and finishing. Moulding and casting shop can help customers with their projects by providing expert advice, guidance, and assistance .

Introduction to Patterns, pattern allowances, ingredients of moulding sand and melting furnaces. Foundry tools and their purposes Demo of mould preparation and Aluminum casting Practice – Study and Preparation of mould for Plastic

- A **pattern** is a replica of the object to be cast, used to prepare the cavity into which molten material will be poured during the casting process.
- **Pattern allowances** are the adjustments made to the pattern to compensate for various factors such as shrinkage, machining, distortion and draft.
- **Moulding sand** is the material used to make the mould cavity. It consists of four main ingredients: silica sand, clay, water and additives. The properties of moulding sand depend on the type and amount of each ingredient.
- **Melting furnaces** are the devices used to heat and melt the metal or alloy to be cast. There are different types of melting furnaces depending

on the fuel, design and capacity. Some examples are cupola, electric arc, induction and crucible furnaces.

- **Foundry tools** are the instruments used to perform various tasks in the foundry process, such as moulding, melting, pouring and finishing. Some common foundry tools are rammer, trowel, sprue cutter, gate cutter, vent wire, ladle, skimmer and chisel.
- **Demo of mould preparation and aluminum casting** is a practical exercise to demonstrate the steps involved in making a mould and casting an aluminum object. The steps are as follows:
  - Select a suitable pattern and apply pattern allowances.
  - Prepare the moulding sand by mixing the ingredients and tempering the sand.
  - Fill a drag (bottom part of the moulding box) with sand and ram it around the pattern.
  - Turn over the drag and place a cope (top part of the moulding box) over it. Align the two parts with dowels.
  - Cut a sprue (vertical channel for pouring molten metal) and a runner (horizontal channel for distributing molten metal) in the cope. Cut a riser (vertical channel for releasing gases and excess metal) in the drag.
  - Vent the mould cavity by poking holes with a vent wire. Remove the pattern carefully from the mould cavity.
  - Heat and melt the aluminum in a crucible furnace. Skim off the impurities from the molten metal.
  - Pour the molten aluminum into the mould cavity through the sprue. Wait for the metal to solidify and cool down.
  - Break the mould and remove the casting. Cut off the sprue, runner and riser. Clean and finish the casting as required.
- **Practice – Study and Preparation of mould for Plastic** is a practical exercise to learn how to make a mould for plastic injection molding. The steps are as follows:
  - Design a 3D model of the plastic part to be produced and generate a 2D drawing with dimensions and tolerances.
  - Select a suitable material and type of mould for the plastic part. Consider the shape, size, complexity and quantity of the part, as well as the properties and cost of the material and the mould.
  - Divide the 3D model into two halves: a core (inner part of the mould) and a cavity (outer part of the mould). Add features such as parting line, draft angle, ejector pins, cooling channels and runner system to the mould design.
  - Fabricate the core and cavity of the mould using CNC machining, EDM, or other methods. Assemble the mould components and test the mould for fit and function.
  - Mount the mould on the injection molding machine and set the parameters such as temperature, pressure, speed and cycle time. Inject the molten plastic into the mould and eject the plastic part. Inspect

the part for quality and defects.

7 CNC Shop 6 Hrs

- CNC stands for Computer Numerical Control, which is a process of using computers to control machine tools such as lathes, mills, routers, and grinders.
- CNC machines can perform various operations such as drilling, cutting, shaping, and engraving on different materials such as metal, wood, plastic, and composite.
- CNC machines are programmed using a code called G-code, which specifies the coordinates, feed rate, spindle speed, and tool selection for each movement of the machine.
- CNC machines can produce high-quality and precise parts with less waste, less human error, and less manual labor than conventional machines.
- CNC machines can also be integrated with other technologies such as CAD (Computer-Aided Design), CAM (Computer-Aided Manufacturing), and CMM (Coordinate Measuring Machine) to optimize the design, production, and inspection of the parts.
- CNC machines require regular maintenance, calibration, and troubleshooting to ensure their optimal performance and safety.
- CNC machines can be classified into different types based on their configuration, such as horizontal, vertical, gantry, and turret. Each type has its own advantages and disadvantages depending on the application and the size and shape of the workpiece.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

- A CNC machine is a computer-controlled machine tool that can perform various operations on a workpiece according to a set of instructions or a program.

- CNC stands for Computer Numerical Control, which means that the machine can convert numerical data into motion commands for the machine tool.

- CNC machines can produce parts with high accuracy, precision, speed, and flexibility, and can operate for long hours without human intervention.

- The main features and working parts of a CNC machine are:

  - Input device: This is the device that is used to input the part program in the CNC machine. The part program contains the instructions for the machine tool to perform the desired operations on the workpiece. The input device can be a punch tape reader, a magnetic tape reader, or a computer via RS-232-C communication .

- Machine control unit (MCU): This is the heart of the CNC machine. It consists of a microprocessor, a memory, and an interface circuit. The MCU receives the part program from the input device, interprets it, and generates the motion commands for the machine tool. The MCU also monitors and controls the speed, position, and feedback of the machine tool .
- Machine tool: This is the part of the CNC machine that performs the actual cutting, drilling, milling, or turning operations on the workpiece. The machine tool has a sliding table and a spindle to control the position and speed of the cutting tool. The machine tool can be a CNC mill, a CNC lathe, a CNC router, a CNC plasma cutter, or any other type of CNC machine .
- Driving system: This is the system that provides the power and motion to the machine tool. The driving system consists of motors, amplifiers, and ball screws or linear guides. The motors can be stepper motors or servo motors, depending on the type and accuracy of the CNC machine. The amplifiers convert the low-voltage signals from the MCU into high-voltage signals for the motors. The ball screws or linear guides convert the rotary motion of the motors into linear motion of the table or the spindle .
- Feedback system: This is the system that measures and reports the actual position and speed of the machine tool to the MCU. The feedback system consists of sensors, encoders, and transducers. The sensors detect the position and speed of the table or the spindle. The encoders convert the analog signals from the sensors into digital signals for the MCU. The transducers convert the physical quantities such as force, torque, temperature, or vibration into electrical signals for the MCU. The feedback system ensures that the machine tool follows the motion commands accurately and compensates for any errors or disturbances .
- Display unit: This is the device that shows the status and information of the CNC machine to the operator. The display unit can be a monitor, a keyboard, a mouse, or a touch screen. The display unit allows the operator to enter or edit the part program, monitor the machine tool, and control the CNC machine .
- Bed: This is the base or the frame of the CNC machine that supports and aligns the machine tool and the workpiece. The bed is made of rigid and heavy material such as cast iron or steel to provide stability and reduce vibration. The bed also has guide ways or rails to allow the movement of the table or the spindle .
- Headstock: This is the part of the CNC machine that holds and rotates the spindle and the cutting tool. The headstock is mounted on the bed and can be fixed or movable, depending on the type of the CNC machine. The headstock also has a motor, a gearbox, and a chuck or a collet to provide power and speed to the spindle and the cutting tool .

- The accessories that can be used with a CNC machine are:
  - Coolant system: This is the system that provides a liquid or a gas to cool and lubricate the cutting tool and the workpiece during the machining process. The coolant system prevents overheating, reduces friction, improves surface finish, and extends tool life. The coolant system consists of a pump, a tank, a nozzle, and a hose. The coolant can be water, oil, air, or mist, depending on

8 To prepare a product using 3D printing 3 Hrs

- 3D printing is a process of creating a three-dimensional object from a digital model by depositing successive layers of material on top of each other.
- To prepare a product using 3D printing, the following steps are required:
  1. Design the product using a computer-aided design (CAD) software or scan an existing object using a 3D scanner.
  2. Convert the design into a 3D printable format, such as STL, OBJ, or AMF, using a slicer software that divides the model into thin layers and generates instructions for the 3D printer.
  3. Choose the appropriate 3D printing technology, material, and settings for the product, depending on the desired properties, quality, and cost. Some of the common 3D printing technologies are fused deposition modeling (FDM), stereolithography (SLA), selective laser sintering (SLS), and direct metal laser sintering (DMLS).
  4. Load the 3D printable file and the material into the 3D printer and start the printing process. The 3D printer will follow the instructions from the slicer software and deposit the material layer by layer until the product is completed.
  5. Remove the product from the 3D printer and perform any post-processing steps, such as removing supports, sanding, polishing, painting, or curing, to improve the appearance and functionality of the product.
- Some of the advantages of 3D printing are:
  - It allows for rapid prototyping and customization of products.
  - It reduces material waste and environmental impact compared to traditional manufacturing methods.
  - It enables the creation of complex and intricate shapes and structures that are difficult or impossible to produce by other means.
  - It lowers the barriers to entry and innovation for individuals and small businesses.
- Some of the challenges and limitations of 3D printing are:
  - It requires a high level of skill and knowledge to design and print

quality products.
  - It can be expensive and time-consuming depending on the size, complexity, and resolution of the product.
  - It may pose ethical, legal, and social issues regarding the ownership, regulation, and safety of the products and the materials used.
  - It may not be able to match the strength, durability, and functionality of products made by conventional methods.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of Total 33 Hrs. I assume you are referring to the book by P. J. Leonard, which is a science fiction thriller about a group of people who wake up in a bunker with only 33 hours of oxygen left. Is that correct? If so, here are some points you can use to study the book:

- The book is divided into three parts: The Bunker, The Surface, and The Truth.
- The main characters are: Jake, a journalist who was investigating a conspiracy theory; Mia, a nurse who was kidnapped by a masked man; Ben, a hacker who was working for a mysterious organization; and Zoe, a scientist who was researching a deadly virus.
- The book explores themes such as survival, trust, morality, and the power of information.
- The book uses flashbacks, multiple perspectives, and cliffhangers to create suspense and mystery.
- The book raises questions such as: Who put them in the bunker and why? What is the connection between them and the virus? What is the truth behind the conspiracy theory? What will they do if they escape the bunker?
- The book has a twist ending that reveals the true identity of the masked man and the motive behind his actions.

Course Outcome:

- A course outcome is a statement that describes what students should be able to do or demonstrate by the end of a course.
- Course outcomes are usually derived from the course objectives, which are the broad goals or purposes of the course.
- Course outcomes should be specific, measurable, achievable, relevant, and time-bound (SMART).
- Course outcomes should align with the course content, activities, assessments, and learning outcomes of the program or department.
- Course outcomes should be communicated to the students at the beginning of the course and throughout the course.
- Course outcomes should be evaluated and revised periodically based on feedback from students, instructors, and other stakeholders.

The students will be able to

- CO1 Use various engineering materials, tools, machines and measuring equipments. K3
  - Identify and classify different types of engineering materials, such as metals, ceramics, polymers, composites, etc.
  - Explain the properties and applications of engineering materials, such as strength, hardness, ductility, conductivity, etc.
  - Select appropriate engineering materials for a given design problem or requirement.
  - Demonstrate the use of various tools, machines and measuring equipments, such as hammers, saws, drills, lathes, CNC machines, vernier calipers, micrometers, etc.
  - Perform basic operations and maintenance of tools, machines and measuring equipments, such as sharpening, lubricating, calibrating, etc.
- CO2 Perform machine operations in lathe and CNC machine. K3
  - Explain the working principle and components of a lathe and a CNC machine.
  - Identify and select the suitable cutting tools, work holding devices, and machining parameters for a given lathe or CNC operation.
  - Perform various lathe operations, such as facing, turning, taper turning, threading, knurling, etc.
  - Perform various CNC operations, such as drilling, milling, contouring, pocketing, etc.
  - Write and execute simple CNC programs using G and M codes.
  - Measure and inspect the machined parts for accuracy and quality.
- CO3 Perform manufacturing operations on components in fitting and carpentry shop. K3
  - Explain the safety rules and precautions to be followed in fitting and carpentry shop.
  - Identify and use the appropriate hand tools, power tools, and measuring instruments for fitting and carpentry work.
  - Perform various fitting operations, such as filing, sawing, chiseling, drilling, tapping, etc.
  - Perform various carpentry operations, such as planing, marking, cutting, joining, etc.
  - Make simple components or joints by fitting and carpentry work, such as vee block, square block, dovetail joint, mortise and tenon joint, etc.
- CO4 Perform operations in welding, moulding, casting and gas cutting. K3
  - Explain the principles and types of welding, moulding, casting and gas cutting processes.
  - Identify and select the suitable welding electrodes, moulding sand, casting metal, and gas cutting equipment for a given operation.
  - Perform various welding operations, such as arc welding, gas welding, spot welding, etc.

- Perform various moulding operations, such as pattern making, sand preparation, mould making, core making, etc.
    - Perform various casting operations, such as melting, pouring, solidification, finishing, etc.
    - Perform various gas cutting operations, such as straight cutting, bevel cutting, circular cutting, etc.
    - Test and inspect the welded, moulded, casted and gas cut parts for defects and quality.
- CO5 Fabricate a job by 3D printing manufacturing technique K3
    - Explain the concept and advantages of 3D printing manufacturing technique.
    - Identify and select the suitable 3D printing material, method, and machine for a given fabrication job.
    - Design and model a 3D object using CAD software, such as SolidWorks, AutoCAD, etc.
    - Convert the 3D model into a printable file format, such as STL, OBJ, etc.
    - Set up and operate the 3D printer, such as loading the material, adjusting the parameters, etc.
    - Print the 3D object layer by layer using the 3D printer.
    - Remove the support material and post-process the 3D printed object, such as cleaning, sanding, painting, etc.

Reference Books:

- Reference books are books that provide authoritative and reliable information on a specific topic or field of study.
- Reference books can be used to find facts, definitions, statistics, biographies, maps, timelines, illustrations, and other relevant data.
- Reference books are usually not meant to be read cover to cover, but rather to be consulted for specific information or purposes.
- Reference books are often organized alphabetically, chronologically, or thematically, and have indexes, tables of contents, glossaries, and bibliographies to help users locate the information they need.
- Some examples of reference books are dictionaries, encyclopedias, almanacs, atlases, directories, handbooks, manuals, guides, and yearbooks.

1. Workshop Practice, H S Bawa, McGraw Hill

- This is a book that covers the basic concepts and skills of workshop technology for engineering students.
- The book is divided into 16 chapters, each focusing on a different aspect of workshop practice, such as safety, tools, materials, machining, welding, fitting, carpentry, etc.
- The book provides theoretical explanations, practical examples, diagrams, illustrations, and exercises to help the students understand and apply the

workshop techniques.
- The book also includes a glossary of technical terms, multiple choice questions, and review questions at the end of each chapter.
- The book is suitable for students of mechanical, civil, electrical, and production engineering, as well as diploma and vocational courses.

2. Mechanical Workshop Practice, K C John, PHI

- This book is a comprehensive guide to the basic principles and practices of mechanical workshop.
- It covers the following topics:
  - Introduction to workshop tools, equipment, and safety measures.
  - Measurement and layout techniques, including linear, angular, and geometric measurements, and use of measuring instruments and gauges.
  - Fitting and assembly operations, such as filing, sawing, drilling, tapping, reaming, and riveting.
  - Sheet metal work, such as cutting, bending, forming, soldering, and brazing.
  - Carpentry and pattern making, such as wood selection, cutting, joining, and finishing, and making patterns for casting.
  - Foundry and forging, such as molding, core making, casting, and defects, and forging operations, tools, and equipment.
  - Welding and gas cutting, such as types, methods, and applications of welding, and oxy-acetylene gas cutting and brazing.
  - Machine shop practice, such as lathe, shaper, planer, milling, drilling, and grinding machines, and their operations and accessories.
  - Electrical and electronics workshop, such as basic electrical and electronic components, circuits, and devices, and their testing and troubleshooting.
- The book is written in a simple and lucid language, with numerous illustrations, diagrams, tables, and examples.
- The book is suitable for students of engineering, diploma, and vocational courses, as well as for hobbyists and professionals.

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications

- Workshop Practice is a course that introduces the students to the basic concepts and skills of various workshop processes, such as fitting, sheet metal, soldering, welding, carpentry, etc.
- Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications are two books that cover the theory and practice of workshop technology in a comprehensive and systematic manner.
- Workshop Practice Vol 1 covers the topics of manufacturing processes, such as casting, forging, metal cutting, machining, welding, etc. It also

includes the study of tools, materials, measurements, and quality control.
- Workshop Practice Vol 2 covers the topics of machine tools, such as lathe, drilling, milling, shaping, grinding, etc. It also includes the study of machine tool accessories, jigs and fixtures, and CNC machines.
- The books are written by S.K. Hajra Choudhury, a renowned author and professor of mechanical engineering, and his co-authors A.K. Hajra Choudhury and Nirjhar Roy.
- The books are published by Media Promoters and Publications, a leading publisher of engineering and technical books in India.
- The books are suitable for undergraduate students of mechanical engineering and related disciplines, as well as for diploma and ITI students, and professionals in the field of workshop technology.

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication

- This text-book explains the fundamentals of NC/CNC machine tools, operations and part programming which form essential portion of course on Computer Aided Manufacturing (CAM) .
- This book also covers advanced topics such as Macro programming, DNC and Computer Aided Part Programming (CAPP) in detail .
- The book is divided into 12 chapters, which are as follows:
  - Chapter 1: Introduction to NC/CNC Machine Tools
  - Chapter 2: Fundamentals of NC/CNC Machine Tools
  - Chapter 3: NC/CNC Machine Tool Systems
  - Chapter 4: NC/CNC Machine Tool Operations
  - Chapter 5: NC/CNC Part Programming
  - Chapter 6: Manual Part Programming
  - Chapter 7: Computer Aided Part Programming (CAPP)
  - Chapter 8: Macro Programming
  - Chapter 9: Direct Numerical Control (DNC)
  - Chapter 10: NC/CNC Machine Tool Selection and Application
  - Chapter 11: NC/CNC Machine Tool Maintenance and Troubleshooting
  - Chapter 12: NC/CNC Machine Tool Simulation and Verification
- The book is written in a simple and lucid language, with numerous examples, diagrams, tables and exercises to enhance the understanding of the concepts and applications .
- The book is suitable for undergraduate and postgraduate students of mechanical engineering, production engineering, industrial engineering and other related disciplines. It is also useful for practicing engineers, technicians and managers involved in the field of NC/CNC machine tools and manufacturing .